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The Heisenberg Cut as a Physical Threshold: Location, Width, and Consequences of the Classical–Quantum Boundary in Detector Dynamics

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Abstract. The Heisenberg cut — the point at which the von Neumann measurement chain is severed and unitary description gives way to definite outcomes — is standardly a bookkeeping convention: movable, unphysical, chosen for convenience, and famously objectionable for exactly that reason. We argue that the cut is a physical threshold with a computable location and a computable width. The argument imports one theorem from a companion paper on outcome selection: below the registration threshold a driven absorber mode has no autonomous phase — it is a globally contracted linear system, its phase slaved to the drive, returning its off-shell excitation to the common field at the rate $\kappa_{\text{ret}} = \Delta E/\hbar$ (a physically motivated ansatz whose microscopic derivation remains open) — while above threshold the mode owns a phase that can lock. The cut sits where the phase-restoring rate falls to the coupling, $\kappa_{\text{ret}} \sim K$, and its fractional width is $w = K/\omega$: sharp beyond resolution for atomic systems ($w \sim 10^{-8}$ – 10^{-6}), percent-level for broadband solid-state absorbers ($w \sim 10^{-2}$). On this account the celebrated movability of the conventional cut is

explained rather than denied — within the reversible sector the placement of the system–apparatus split is provably free, and that freedom hardens into physics only at the locking layer. The framework cleanly separates three notions that measurement talk chronically conflates — entangling interaction, decoherence, and the nonlinear lock — and locates the central interpretive dispute in the third. Consequences follow for the anatomy of a measurement event (capture, selection, registration, with reversibility ending precisely at the layer), for the virtual/real distinction (a heuristic correspondence: off-shell excitation as a failed lock whose lifetime is the energy–time uncertainty; on-shell as locked, with critical slowing at the boundary), and for experiment: the cut has an address in every laboratory, the layer width is a dial that existing platforms already span, and the threshold picture makes coupling — not mass, size, or gravitational self-energy — the variable that decides where quantum behavior ends, a discriminator against both spontaneous-collapse and gravitational-reduction programs. All claims are conditional on the companion paper’s single-detector premises; nothing here consumes its nonlocal ontology.

1. Introduction

Quantum mechanics describes a measurement twice. Inside the formalism, apparatus and system entangle unitarily and nothing ever happens; outside it, an experimenter reads one number off one dial. Somewhere between the two descriptions a line is drawn — system on one side, classical apparatus on the other — and the theory’s predictions are indifferent to where. Heisenberg noted the indifference; von Neumann proved a version of it, showing the chain system $\rightarrow$ detector $\rightarrow$ amplifier $\rightarrow$ observer can be cut at any link with identical statistics [von Neumann 1932]; Bohr elevated the line to a principle while declining to locate it. Bell, sixty years later, made the objection canonical: a “shifty split” is no foundation for a physical theory — nothing in physics should depend on a boundary that is nowhere in particular [“Against ‘measurement’”: Bell 1990].

The modern consensus assigns the explanatory work to decoherence: interaction with an environment diagonalizes the reduced density matrix in a preferred pointer basis on absurdly short timescales, and the quantum/classical transition is thereby “explained” [Zeh 1970; Zurek 1981; Zurek 1982; Joos & Zeh 1985; Zurek 2003; Schlosshauer 2004]. The decoherence literature itself is admirably candid that this answers a different question. Diagonality is not an event; a mixture improper for the universe as a whole selects no member of itself; and the program’s own reviews state plainly that decoherence does not solve the problem of outcomes [Schlosshauer 2004; Adler 2003]. What decoherence explains is why *interference* becomes unobservable — why the density matrix commutes with the pointer observable — not why *this* pointer position occurs. The cut therefore survives the decoherence program intact: it has merely been renamed the point at which one stops speaking of improper mixtures and starts speaking of occurred events.

This paper claims the cut is physical, locatable, and has a width. The claim is not free-standing: it imports its machinery from a companion paper [Paper 1], which develops outcome selection as a stochastic energy game among detector absorber sites and proves, from stated premises about detectors, that the game is fair and its statistics Born. One theorem of that paper does the work here. **Theorem 2** (restated in §3): an absorber holding energy $e < E_0$ is off-shell by $\Delta E = E_0 - e$ and constitutes a driven *linear* mode returning its excitation at the rate $\kappa_{\text{ret}} = \Delta E/\hbar$; it is globally contracted onto a fixed point whose phase is slaved to the drive; it possesses no zero mode, no autonomous clock, and no capacity for the phase-coherent feedback that irreversible registration requires. Only within a boundary layer $\Delta E \lesssim \hbar K$ of the shell — fractional width $w = K/\omega$, with K the site–field coupling — does an autonomous phase emerge and locking become possible. Far from the shell ($\Delta E \gg \hbar K$): dynamics, interference, reversibility. Inside the layer and beyond it, at the completed lock: commitment. The cut is the layer. (Directional convention, used throughout: “far from the shell” means large deficit ΔE — deeply sub-threshold, deeply reversible; “at the shell” means $\Delta E \rightarrow 0$ — the lock’s doorstep.)

Three things must be kept distinct, and the paper’s architecture follows the distinction. **Settled, and not ours to claim:** the unitary indifference to the placement of the system–apparatus split (von Neumann’s theorem, restated in §6 as the *reason* the conventional cut is movable); decoherence theory in full — einselection, pointer bases, the suppression of interference, with its own conclusion that no outcome is thereby

selected; and the demonstration by Allahverdyan, Balian and Nieuwenhuizen that *registration* — the amplification of a selected microscopic asymmetry into a stable macroscopic record — is ordinary statistical mechanics of a metastable apparatus [Allahverdyan et al. 2013]. **Claimed:** that the three-way distinction of §2 (entangling interaction / decoherence / nonlinear lock) is physically real; that the lock supplies exactly the element the settled results lack; that the cut is located at $\kappa_{\text{ret}} \sim K$ with width $w = K/\omega$, computable system by system; and that this yields a detector taxonomy, a reading of the virtual/real distinction, and experimental discriminators. **Open:** whether the sub-threshold slaved-phase statement survives full field quantization (inherited from Paper 1’s open problems); the closed dynamical model of the lock itself; and everything involving the entangled sector, which this paper does not touch.

That last point deserves emphasis, because it makes this paper’s conditionality strictly weaker than its companion’s. Paper 1’s premises divide into a single-detector core (wave realism P0; capture energetics P1; amplitude-linear coupling P2; detector initial conditions P3; registration taxonomy P4) and a nonlocal extension (shared registry P5; ordering foliation P6) consumed only by its multi-quantum and entangled-pair constructions. The present paper consumes the core alone. A reader who rejects the nonlocal ontology outright can still evaluate everything here.

The paper proceeds as follows. §2 states the three-way distinction. §3 presents the threshold quantitatively and tabulates the layer width across laboratory systems. §4 maps the anatomy of a measurement event onto the threshold and derives the detector taxonomy. §5 reads the virtual/real distinction as sub-/supra-threshold. §6 explains the movable cut — deriving both its freedom and the boundary of that freedom. §7 states the single-world picture and prices it against the alternatives. §8 assembles the experimental consequences. §9 situates the result and lists open problems.

2. Three things not to conflate

Measurement talk chronically merges three physically distinct processes. The central disputes about the measurement problem, in our reading, are disputes about the third; clarity about the first two is what the settled literature already provides.

(1) Entangling interaction. Couple system to apparatus and let them interact. On the joint Hilbert space this is plain unitary evolution: $i\hbar \partial_t \Psi = H_{\text{tot}} \Psi$. Entanglement grows; the map is linear and *reversible*. Nothing here distinguishes a measurement from any other interaction — which is precisely von Neumann’s point, and the origin of the chain.

(2) Decoherence. Open the apparatus to a large environment and trace it out. The reduced state obeys a completely positive, irreversible master equation — but one *still linear in the density matrix*. Off-diagonal terms in the pointer basis are suppressed at spectacular rates; the state comes to *look* like a classical mixture. But linearity is decisive: a linear map cannot send a superposition to one of its branches, only to a mixture of them, weighted exactly as before. Irreversibility has been purchased; selection has not. Dissipation is not nonlinearity.

(3) The lock. The genuinely nonlinear step: a symmetry-breaking dynamics that takes the distributed, decohered excitation and commits it to one basin — one site, one port, one pointer position. This is where an event becomes an event. Paper 1’s content is that the lock exists as detector physics, that its statistics are provably Born under stated premises, and that the noise driving it is the ordinary interference of the site’s amplitude with the uncontrolled-phase fields at the detector surface. The present paper’s content is what the lock’s *threshold structure* implies for the geography of the quantum–classical boundary.

In this language the measurement problem decomposes cleanly. Stage (1) is settled unitary dynamics; stage (2) is settled open-system dynamics; the interpretations disagree about stage (3). Everett denies it exists (all basins are occupied); GRW/CSL postulate it as new fundamental stochasticity with new constants [Ghirardi et al. 1986; Bassi & Ghirardi 2003]; Penrose supplies a threshold for it without a mechanism [Penrose 1996]; decoherence-only accounts stop at (2) and redescribe the question. The claim defended here: (3) is ordinary — if nonlinear — detector dynamics, and its threshold is the cut.

The three-way distinction also disciplines vocabulary that this paper will need. *Reversible* applies to stages (1) and, in the echo/rephasing sense, to the capture stage of §4 — evolution whose effects can be undone

by feasible operations. *Irreversible-but-linear* is stage (2): information delocalized beyond practical recovery, statistics unchanged. *Committed* is stage (3) alone. The cut, properly speaking, is the boundary of commitment — not of reversibility (decoherence already ends practical reversibility) and not of interaction (everything interacts). Locating the cut therefore means locating the onset of the nonlinearity.

3. The threshold, quantitatively

3.1 The slaved phase and the layer

We restate the companion result in the form used throughout ([Paper 1] Theorem 2 and Appendix B; proof sketch below, so that no foundational claim rests on an unavailable text). An absorber site short of the full quantum is off-shell: it holds $e < E_0$, with deficit $\Delta E = E_0 - e$, and P4(i) — no stationary state below the transition — makes that deficit a *return rate*, $\kappa_{\text{ret}} = \Delta E/\hbar$: the rate at which the virtual (unbankable) excitation reversibly returns to the common field, the beat rate of a failed lock. Two things about this rate, stated before it is used. First, it is not bath dissipation — the returned energy re-enters the exchange pool the selection game redistributes; the symbols γ, Γ are reserved in this paper for genuine dissipative rates. Second, its form is a *physically motivated ansatz*: that the return rate equals $\Delta E/\hbar$, and that reversible give-back may be modeled as a contraction eigenvalue in the site’s reduced description, await the microscopic derivation logged in §9 and in the companion’s open problems ([Paper 1] §9.4(viii)). Everything below is conditional on the ansatz surviving that calculation — throughout this paper, read “if the mean-field absorber model survives quantization” in front of every quantitative claim. In the rotating frame, with amplitude $a = re^{i\varphi}$, detuning Δ (the drive–transition frequency mismatch, an independent parameter), and drive $fe^{i\varphi_d}$:

$$\dot{r} = -\kappa_{\text{ret}} r + f \cos(\varphi_d - \varphi), \quad \dot{\varphi} = \Delta + \frac{f}{r} \sin(\varphi_d - \varphi).$$

In Cartesian coordinates this system is *linear*, with eigenvalues $\lambda = -\kappa_{\text{ret}} \pm i\Delta$: a unique, globally attracting fixed point, phase offset $\varphi^* = \varphi_d - \arctan(\Delta/\kappa_{\text{ret}})$ determined by $(\kappa_{\text{ret}}, \Delta)$ alone — identical for identical sites, independent of stored energy. The phase is an enslaved coordinate. No trajectory winds; perturbations decay at rate κ_{ret} ; there are no running solutions and hence no entrainment pull. The site has no clock of its own. (Proof sketch of the companion theorem: linearity in Cartesian coordinates makes the fixed point’s global attraction an eigenvalue statement; the absence of running solutions excludes the Adler entrainment average, which is derived from winding trajectories of an *autonomous* limit-cycle oscillator; and P4(i)’s level discreteness prevents banking partial winnings, enforcing threshold sharpness. Full statement and supporting results: [Paper 1] §5.2, Appendix B.)

Dictionary (maintained throughout, and in the companion from its v0.5.3). $\kappa_{\text{ret}} = \Delta E/\hbar$: deficit-induced reversible return rate — state-dependent, vanishing at the shell. Δ : drive–transition detuning. K : coherent site–field coupling — the quantity that sets the layer width $w = K/\omega$. Γ : a *measured* coherence-decay rate (linewidth, $1/T_2$, dephasing), a genuinely dissipative quantity used in §3.3 as a proxy for K under stated conditions. $\Delta\omega$: the detuning variable of §5’s Adler analysis. Five symbols, five different quantities; the reviewed draft’s conflations of them drew justified fire, and the distinctions are load-bearing.

We flag at once the sharpest open formulation, owed to the review panel: the sub-threshold equation above contains no K — the comparison $\kappa_{\text{ret}} \lesssim K$ sets a scale, but no single continuous dynamical model yet spans both sides of the boundary and exhibits the bifurcation at $\kappa_{\text{ret}}/K = 1$ with coefficients derived from absorber microphysics. Constructing that model *is* the open problem (§9), and the classical oscillator of Figure 1 is an analogy for what its solution should look like, not a substitute for it.

Contrast the autonomous oscillator presupposed by the Adler equation $\dot{\varphi} = \Delta - K \sin \varphi$ [Adler 1946]: amplitude held on a limit cycle by the oscillator’s own gain, phase a neutrally stable zero mode, free to run — and, when running, exerting the nonzero average pull that is the engine of injection locking, laser mode competition, and every rich-get-richer instability in synchronization physics [Pikovsky et al. 2001]. The qualitative difference between the two regimes is the theorem: *below threshold there is no zero mode to entrain*. A free clock — the prerequisite for systematic, phase-coherent energy extraction, and hence for winner-take-all feedback and for irreversible commitment — emerges only where the restoring rate collapses,

$\kappa_{\text{ret}} \lesssim K$, i.e. within a boundary layer

$$\Delta E \lesssim \hbar K \quad \Longleftrightarrow \quad w \equiv \frac{\Delta E_{\text{layer}}}{E_0} = \frac{K}{\omega}$$

of the shell. Figure 1(a) displays the transition as a single observable: an injected oscillator’s mean phase-winding rate, swept through threshold. Below, exactly zero — the drive owns the phase. Above, the free clock switches on and runs at the autonomous Adler rate. The crossover is the layer.

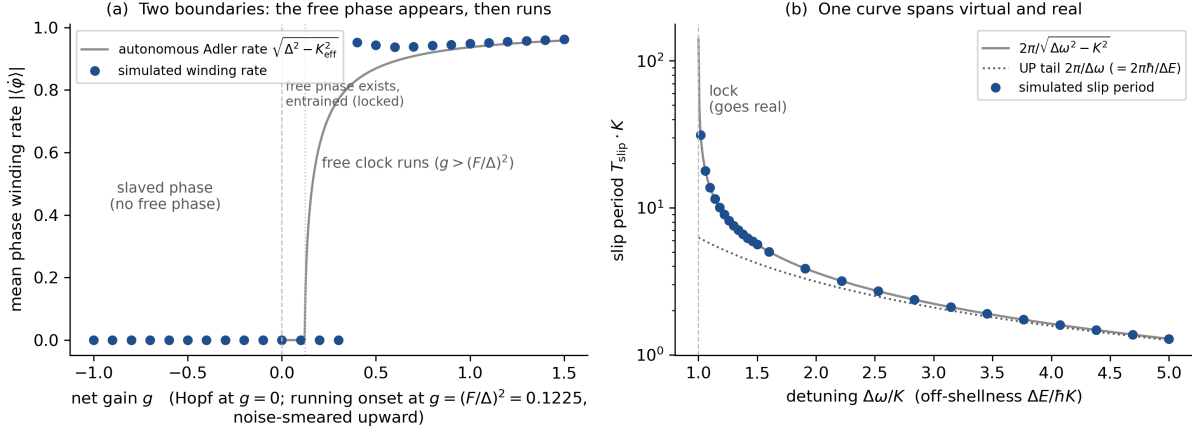


Figure 1: Schematic oscillator analogy — not a simulation of a detector. **(a)** Injected Stuart–Landau oscillator, net gain g swept at fixed detuning Δ and injection F . Two distinct boundaries, both marked: at the Hopf threshold $g = 0$ the free phase comes to *exist* (the linear fixed point gives way to a limit cycle); the winding observable turns on only at the *injection-locking boundary* $g = (F/\Delta)^2 = 0.1225$ (noise-smeared upward, onset near $g \approx 0.35$ at these parameters), where the free clock can no longer be entrained and *runs*, following the autonomous Adler rate $\sqrt{\Delta^2 - K_{\text{eff}}^2}$. The reviewed draft’s caption conflated these two thresholds; the correction is owed to the review panel’s independent re-run of this script. **(b)** One curve spans the §5 correspondence: Adler slip period versus detuning. Far off-resonance the slip period is the energy–time-uncertainty lifetime $2\pi\hbar/\Delta E$ (dotted); at the tongue boundary it diverges — critical slowing — and the mode locks. Simulated points; analytic curves; methods, convergence, and regeneration command in Appendix A.

3.2 The width, two ways

The primary result is $w = K/\omega$, from marginality: the phase becomes exploitable when its restoring rate falls below the coupling, $\kappa_{\text{ret}} \lesssim K$. The companion’s second route — the width–commitment identity, $K \sim \Gamma$ by a fluctuation–dissipation argument, giving $w = \Gamma/\omega$ — is, as the review panel correctly insisted, an *identification, not an identity*: coherent coupling and coherence decay are model-dependently related, with prefactors and spectral-density structure the argument does not supply, and they can be *independently* engineered — cavity QED tunes K by geometry and detuning while holding damping fixed, which is a standing counterexample to any universal $K \sim \Gamma$. The scope of the proxy is therefore stated explicitly: **for passive absorbers** — sites whose coupling to the common field is the same interaction that sets their radiative/dephasing linewidth — $K \sim \Gamma$ is a defensible order-of-magnitude default, and it is what makes the layer width *empirically tabulable* from measured linewidths and frequencies. Where K is engineered independently, the primary formula $w = K/\omega$ applies with the engineered K , and the proxy is dropped. Deriving the precise prefactor for one concrete absorber is a named open task (§9).

3.3 The cut has an address: the layer width across laboratory systems

All rates are angular ($\Gamma = 1/\tau$ for a lifetime τ ; linewidths quoted in Hz converted by 2π); each row is anchored to the cited source. **The table is order-of-magnitude, under the passive-absorber proxy $K \sim \Gamma$ of §3.2** — its entries are measured coherence scales converted to layer widths under that stated assumption, not model-independent measurements of the cut.

System	Γ (s^{-1})	ω (s^{-1})	$w = \Gamma/\omega$	character of the cut
Alkali D-line, natural width [Steck 2025]	$3\text{--}6 \times 10^7$	$2.1\text{--}3.2 \times 10^{15}$	$\sim 1.5 \times 10^{-8}$	razor-sharp; Born exact to all practical resolution
Optical clock transition (^{87}Sr , Al^+) [Dolde et al. 2025; Ludlow et al. 2015]	$4 \times 10^{-3}\text{--}5 \times 10^{-2}$	$2.7\text{--}7.0 \times 10^{15}$	$2\text{--}7 \times 10^{-18}$	sharpest cut in any laboratory
Transmon qubit ($1/T_2$, $T_2 \sim 10\text{--}100 \mu\text{s}$) [Kjaergaard et al. 2020]	$10^4\text{--}10^5$	$2.5\text{--}3.8 \times 10^{10}$	$3 \times 10^{-7}\text{--}4 \times 10^{-6}$	atomic-grade sharpness in a macroscopic circuit
Quantum dot, 4 K (radiative $\rightarrow$ spectral-diffusion-broadened) [Kuhlmann et al. 2015]	$10^9\text{--}10^{11}$	$\approx 2.0 \times 10^{15}$	$6 \times 10^{-7}\text{--}5 \times 10^{-5}$	intermediate; the natural dial (§8.2)
Si photodiode / SPAD, 300 K (carrier dephasing) [Sabbah & Riffe 2002]	$4 \times 10^{12}\text{--}10^{14}$	$2.7\text{--}4.7 \times 10^{15}$	$10^{-3}\text{--}3 \times 10^{-2}$	percent-level layer; deviations live here ([Paper 1] §8)

Which Γ , *stated explicitly*. Each row uses the coherence-decay rate appropriate to the locking analysis — natural (radiative) width for isolated atoms, $1/T_2$ for circuits, momentum-relaxation/dephasing for band carriers — never recombination or readout times. Four caveats travel with the table. (i) The alkali row is the *natural* width; Doppler broadening in room-temperature vapor raises the effective Γ by $\sim 10^2$, toward the $w \sim 10^{-6}$ end quoted for atomic lines in [Paper 1]. (ii) The clock row uses the directly measured ^{87}Sr 3P_0 natural lifetime ($\tau = 167$ s); practical clock coherence is laser-limited at $w \lesssim 10^{-15}$, still the sharpest cut ever engineered. (iii) The transmon row is typical circa 2020; current best devices exceed $T_2 \sim 300 \mu\text{s}$ –1 ms, pushing $w \rightarrow 10^{-8}$. (iv) The quantum-dot upper bound is slow spectral diffusion — inhomogeneous in time rather than a true homogeneous rate — and the silicon dephasing time (32 fs at $5 \times 10^{18} \text{ cm}^{-3}$, 300 K) is density-dependent.

Two readings of the table. First, the cut is *not* at a universal scale: it sits at a different absolute energy in every system, exactly as a convention-based cut would — but its *location in the dimensionless variable* κ_{ret}/K is universal, and its width is predicted, not chosen. Second, the table dissolves a familiar puzzle: superconducting circuits are “macroscopic” by any particle-count measure, yet they interfere, violate Bell and Leggett–Garg inequalities, and behave as clean two-level systems. On the threshold account this is unmythical: their w is atomic-grade. Macroscopicity in mass or particle number is simply not the variable; κ_{ret}/K is. §8.4 turns this observation into a discriminator.

4. Anatomy of a measurement event

4.1 Three stages, one crossing

The reviewed draft of this section described the lock as both the commitment boundary and as “engaging reversibly,” and the panel correctly identified the collision (finding N3). We repair it with one operational criterion, stated first and used everywhere: **the cut is where in-principle recoverability ends** — the first point in the chain at which no operation, however idealized, can restore the pre-event configuration. Against that criterion, one consistent time-ordering:

this paper (§2)	companion ([Paper 1])	dynamics	recoverability
stage (1) entangling interaction	capture	unitary, distributed excitation	recoverable — demonstrated (echoes, quantum memories)
stage (3) begins	selection: lock <i>engagement</i>	fast synchronization; exchange continues	recoverable in principle — the engaged lock can still dissolve
the cut	selection: lock <i>completion</i>	one site assembles the full quantum (absorption at $s = 1$)	in-principle recoverability ends here
stage (2) + amplification	registration	slow dissipative commit; record written	irrecoverable and now also macroscopic

Capture completes on the fastest timescale the interaction permits ($\tau \gtrsim \hbar/\Delta E$, saturated at $\sim 1/\omega$ for a massless quantum). The lock’s *engagement* is reversible — that is the content of the companion’s temporal ladder, and the word “lock” in the reviewed draft ambiguously named both phases. The lock’s *completion* — the whole stake at one site — is the crossing. Registration, the Curie–Weiss physics of Allahverdyan–Balian–Nieuwenhuizen [Allahverdyan et al. 2013], finalizes and amplifies what completion decided; by the companion’s Theorem 4 (commit-rate independence) its speed never touches the statistics. The commonplace that “collapse happens when the record is amplified” misassigns the cut by one stage: amplification finalizes; completion decides.

Capture’s reversibility, scoped honestly (panel finding N5): it is *demonstrated* for controlled, discrete-level ensembles — spin echoes, photon echoes, and atomic-frequency-comb memories run capture backwards on the bench, at the single-photon level, with no registration having occurred [Hahn 1950; Kurnit et al. 1964; de Riedmatten et al. 2008; Afzelius et al. 2009] — and *in-principle but undemonstrated, and practically foreclosed*, for broadband continuum absorbers, whose reference structure (below) is uncontrolled. The citations establish rephasing and retrieval; the extension of “reversible capture” to arbitrary detector bulks is this framework’s inference, not the cited experiments’ content.

Recoverability as history: the reference trichotomy. [Labeled interpretation, sponsor-originated; developed jointly with the companion’s v0.5.3 §4.] What recoverability physically *is*: a reversal requires the stored phase history to point back along the path — a wavefront can be run backward only by an agent holding its complete phase record, as phase-conjugate holographic readout demonstrates operationally. Whether the record survives is set by the *reference structure* of the medium:

reference structure across sites	record	consequence
common (hologram, homodyne local oscillator, AFC comb)	invertible	reversible; wavefront reconstructable
different but deterministically known (inhomogeneous ensemble)	invertible by rephasing	echo-recoverable — capture without commitment
random and unrecorded (P3(a) detector bulk; each exchange hop)	non-invertible	history diffuses; commitment possible — the cut’s side of the line

The middle row is why echoes work; the reversible/irreversible dephasing split of magnetic resonance is this trichotomy under its textbook name. The bottom row is the companion’s P3(a) — the same reference incoherence that makes the selection game fair makes the record non-invertible, so fairness and irreversibility share one origin. In this language the cut criterion above reads: *the cut is crossed when the incident wave’s history becomes non-invertible* — diluted across unrecorded references faster than any operation could reassemble it — and Zurek’s quantum Darwinism describes the complementary currency: records of *the outcome* proliferate redundantly [Zurek 2003] precisely as records of *the incident wave’s history* die. The crossover between the two record types is another face of the cut.

4.2 The detector taxonomy as threshold phenomenology

Two classifications of real detectors follow from the threshold structure, and their cross-product is exhaustive.

By spectrum (P4). *Discrete-level absorbers* (atoms, molecules, ions): no stationary state below the transition, so the full quantum must assemble at one site; the layer is razor-thin (Table, rows 1–2); sub-threshold excitation is virtual and unbankable — level discreteness enforces threshold sharpness. *Continuum absorbers* (semiconductors, metals): final states at all energies above the gap; registration is a rate process throughout selection; the layer is wide ($w \sim 10^{-2}$) and the percent-level deviation phenomenology of Paper 1 §8 lives in it.

By record structure. *Event detectors* (photomultiplier, SPAD, single atom): one lock, one record — the selection game runs once. *Track-recording media* (cloud chamber, emulsion, silicon tracker): Mott’s problem [Mott 1929] — a single spherical wave produces a straight track — is, in threshold language, a *sequence of partial locks*: each grain along the track runs a selection game whose initial stakes are conditioned by the preceding lock (the wave re-formed around the previous registration), so the sequence is a chain of conditional Born games, each crossing its own layer. Mott’s unitary demonstration that the amplitude concentrates on collinear configurations supplies the conditional stakes; the locks supply the events. A track is not one measurement but a correlated cascade of them — which is why track detectors, not interferometers, are where the classical world’s *trajectories* come from.

The taxonomy earns its keep twice below: it sorts every entry of the deviation phenomenology (§8 inherits Paper 1’s rule that class-(ii) continuum detectors evade late-game bias entirely via commit-rate independence), and it blocks a familiar false generalization — conclusions drawn from event detectors do not automatically transfer to tracking media, whose cascade structure is the reason “watching a trajectory” feels classical.

5. Virtual and real: a heuristic correspondence

(Retitled and rescope in v0.3, per the panel’s convergent finding: what follows is a correspondence between the locking threshold and the virtual/real vocabulary — not a field-theoretic derivation, and no later result rests on it.) The same threshold that draws the cut can be read onto the virtual/real line, and the reading is quantitative in one respect. The dictionary: off-shell energy imbalance ΔE corresponds to detuning $\hbar\Delta\omega$; the lock threshold is the mass shell.

A virtual excitation is an unlocked, off-resonance mode. It slips against the drive and dies as a transient; its lifetime is the slip time of a failed lock. For the Adler phase outside the locking range, the slip period is

$$\tau_{\text{slip}} = \frac{2\pi}{\sqrt{\Delta\omega^2 - K^2}} \xrightarrow{\Delta\omega \gg K} \frac{2\pi}{\Delta\omega} = \frac{2\pi\hbar}{\Delta E},$$

which *is* the energy–time uncertainty lifetime — not postulated but produced, as the beat period of an excitation that fails to lock (Figure 1(b), dotted asymptote). At the tongue boundary $\Delta\omega \rightarrow K^+$ the slip period diverges: critical slowing. The excitation stops slipping, locks, and persists — the correspondence’s reading of “going on-shell.” One curve spans both regimes (Figure 1(b)): the far tail reproduces the uncertainty-principle lifetime; the divergence marks the lock; the boundary between them is the same $\kappa_{\text{ret}} \sim K$ layer that §3 identified as the cut. The governing clock changes at the crossing: $\hbar/\Delta E$ (off-shell borrowing) below; infinite (stable attractor) or $\hbar/\Gamma$ (golden-rule decay through open channels) above. For a real but unstable

particle the product $\Gamma\tau \sim \hbar$ still *resembles* the uncertainty principle, but as a consequence — the Fourier width of an exponentially decaying amplitude — not as the off-shell borrowing that governs the virtual side.

Status of this section, stated plainly — now in the panel’s own terms. In quantum field theory, off-shellness is a statement about four-momenta failing a dispersion relation; virtual excitations are internal contributions to perturbative amplitudes, not observable transients carrying borrowed energy on an uncertainty clock; the energy–time relation is not a license to violate conservation for a prescribed time; and the 2π in the slip-time formula is an artifact of the Adler model, not a QFT theorem. No propagator, spectral-function, pole, or LSZ-type argument connects the locking boundary to a mass shell here — a genuine field-theoretic version would have to derive the detector’s Green function and show its poles or spectral support reorganizing at the locking transition, which no one has done. The correspondence is therefore a *re-description with one quantitative consequence*, not an independent derivation: standard quantum field theory computes with off-shell propagators and needs no story about what a virtual excitation “is doing,” and “virtual-particle lifetime” is itself a heuristic gloss on a propagator integral. What the re-description adds is the middle of Figure 1(b): a definite, measurable crossover profile — the critical-slowness divergence at $\Delta E \rightarrow \hbar K$ — where the standard heuristic offers only the two asymptotes. Systems that can be tuned through the layer (§8.2) can in principle trace the middle of the curve; that, and not the asymptotes, is the falsifiable content. We flag the section’s altitude accordingly and rest no later result on it.

6. The movable cut, explained

6.1 Why the conventional cut is movable: a theorem, not an embarrassment

Within the reversible sector, the placement of the system–apparatus split is provably free. Von Neumann’s chain argument is exactly this: model the detector quantum-mechanically or classically, cut after the system, after the detector, after the amplifier — all choices yield identical statistics, because the intervening dynamics is linear and the eventual outcome weights are basis-quadratic either way [von Neumann 1932]. (Assumptions carried by that argument, stated since we lean on it: the intervening links evolve unitarily; the pointer coupling is of the measurement form that correlates without disturbing the measured basis; and the Born weights are applied, wherever the cut falls, to the same Schmidt coefficients — it is a statistics-equivalence theorem, not an ontology.) The threshold account *keeps this theorem and explains its domain*. Every link of the chain that lies above the layer — every stage whose dynamics is stage-(1) unitary or stage-(2) linear-dissipative — can be placed on either side of the description at zero cost, precisely because linear dynamics commutes with the bookkeeping. The freedom to move the cut is a symptom of linearity, and linearity is exactly what holds *everywhere except the layer*.

The freedom ends where the nonlinearity begins. The lock cannot be placed on the “described unitarily” side of the split, because no unitary (or linear-dissipative) description reproduces commitment: a linear map preserves the superposition’s weights and can only deliver mixtures (§2). The layer is therefore the unique link of the chain at which the two descriptions cease to be intertranslatable — the one place the cut *must* go if the description on each side is to be dynamically accurate. The conventional cut’s shiftiness and the physical cut’s definiteness coexist without tension: the former is the linearity theorem, the latter is the locking threshold, and the confusion between them dissolves once stages (1)–(3) are kept separate.

Bell’s complaint is thereby answered in its own terms. The split is shiftily *within the sector where nothing depends on it* — that is a theorem, and no foundation should be built on resisting a theorem. It is pinned — located at $\kappa_{\text{ret}} \sim K$, with width $w = K/\omega$ — exactly where something *does* depend on it. Nothing fundamental is nowhere in particular.

6.2 The cut as a crossover

A boundary with a dimensionless location and a finite width is, in the language of many-body physics, a *crossover*, and the naming buys discipline. The location is $\kappa_{\text{ret}}/K = 1$; the width is $w = K/\omega$; on one side one set of laws dominates (reversible, interference-preserving, split-placement-free), on the other a different term dominates (dissipative, committing, split-pinning). Nothing discontinuous happens to the laws themselves — the locking term is present in the dynamics throughout, and simply has nothing to grip while the phase

is slaved. A crossover, not a phase transition; a dominance boundary, not a change of legislation.

Two consequences of the naming. First, *dose-response*: observables that probe the boundary should vary smoothly with the crossover variable κ_{ret}/K , with all deviation phenomenology scaling with the layer width — which is exactly the structure of Paper 1’s deviation ledger (deviations $O(w)$: negligible for atomic detectors, percent-level for broadband solids). A conventional cut predicts no such systematics; a crossover mandates them. Second, *completion elsewhere*: in the parent framework the dissipative sector is also where a preferred foliation enters the dynamics, so the same crossover marks where frame-sensitive physics switches from present-but-decoupled to dynamically active. That identification, its symmetry accounting, and its experimental program belong to the framework paper and are deliberately not developed here; we note only that the cut’s crossover structure is what makes such a completion well-formed [Paper 3, in preparation].

7. Single world: the occupied basin

The threshold picture yields a one-world reading of the formalism. Above the layer, the wave explores; superposition is physical, interference is real, and *no basin is occupied* — the selection game has not run. At the layer, symmetry breaks: noise — the surface-interference noise identified in Paper 1 — tips the distributed excitation into one basin, and the system *occupies* it. The other basins do not become other worlds; they become unoccupied attractors, exactly as the untaken well of a buckling beam is not a parallel beam. Where Everett reads the persistence of the other branches’ amplitudes as their reality, the threshold account reads their amplitudes as *stakes that lost* — physically real before the game (P0 wave realism is not negotiable here), physically redistributed by it (the energy went to the winner; Paper 1 §4).

Pricing against the alternatives, with our own costs on the table:

Program	supplies an event?	mechanism?	new constants?	principal cost
Decoherence-only	no	—	no	redescribes the question; cut survives renamed
Everett/MWI	no <i>single</i> outcome (branching; “event” in this paper’s sense)	—	no	all branches real; Born weights via decision theory (its adequacy contested both ways)
GRW/CSL	yes	postulated noise	yes (λ, r_C)	new physics engineered for the job; anomalies predicted [Bassi & Ghirardi 2003; Bassi et al. 2013]
Penrose OR	threshold only	none offered	gravitational scale	location without dynamics [Penrose 1996]
ABN (Curie–Weiss)	registration only	statistical mechanics	no	selection assumed at the crucial step [Allahverdyan et al. 2013]

Program	supplies an event?	mechanism?	new constants?	principal cost
This program	yes	detector locking (Paper 1)	no	conditional on P0–P4 (incl. wave realism); κ_{ret} ansatz and lock dynamics not yet derived; Theorem 2 linear-semiclassical, quantization open

The comparison is deliberately double-edged. Against GRW/CSL: no new constants, no engineered noise — the fairness is proved, not tuned, and the “collapse rate” is the physical commit rate of an actual detector. Against Penrose: a mechanism where he supplies only a scale — though his structural instinct, that the boundary is physical and dynamical rather than conventional, is the same instinct this paper formalizes. Against MWI: one world, at the price of wave realism and the two open problems we carry visibly (the field-quantized sub-threshold statement; the closed lock dynamics). Against decoherence-only: we accept every einselection result wholesale and add the one thing its own reviews concede is missing. And we weight our own column the way the panel insisted it be weighted: “no new constants” is genuine, but wave realism, the κ_{ret} ansatz, the unclosed lock, and (for the entangled sector, elsewhere) a preferred foliation are non-trivial ontological and theoretical costs relative to decoherence-only or Everett — the advantage claimed is not cheapness but *checkability*: this program pays in premises about detectors, which laboratories can interrogate, rather than in metaphysics, which they cannot.

8. Where the cut bites: experimental consequences

8.1 Inherited: the deviation phenomenology scales with the layer

Everything in Paper 1’s deviation ledger is, in the present language, layer phenomenology: deviations from exact Born statistics are $O(w)$, hence invisible for atomic-grade detectors and percent-level where $w \sim 10^{-2}$ (broadband solids), concentrated in configurations that couple to the layer (mismatched ports; pre-cohered absorbers). We do not re-derive that ledger; we re-read it as the dose–response curve of the crossover (§6.2): *the same experiments that test Paper 1’s selection mechanism test this paper’s cut location*, because the deviations are the layer made visible.

8.2 The dial: systems engineered to straddle the layer

The table of §3.3 spans sixteen orders of magnitude in w — from an optical-clock transition at 10^{-18} to a room-temperature photodiode at 10^{-2} — and, decisively, some platforms can *tune* it: a quantum dot’s linewidth varies orders of magnitude with temperature and phonon coupling; a transmon’s dissipation is engineered via its Purcell environment; cavity QED sets K directly by detuning and geometry. The reviewed draft claimed here that boundary phenomenology should vary “with κ_{ret}/K and nothing else” and called it the paper’s primary novel experimental claim. The panel’s objection is accepted in full: stated at that altitude, the claim is close to dimensional analysis on coherence phenomenology, which standard open-system theory also organizes by coupling and linewidth ratios; real devices carry bath structure, drive, temperature, geometry, and disorder as separate axes; and no proposed-theory curve yet exists to compare against a null model. **The one-parameter collapse is therefore a research direction, not a prediction**, pending the crossover-profile calculation of §9.

What replaces it is a protocol sketch, offered in the preregistrable format the panel specified. *Platform*: a transmon coupled to a lossy readout mode — the one system where the two rates are independently engineered (K by coupler geometry/detuning; dissipation by Purcell filter). *Sweep*: hold the qubit and drive fixed; step the engineered coupling so the ratio crosses the putative layer. *Observables*: (i) the distribution of commitment latencies (time from drive arrival to irreversible readout excursion) and (ii) the recoverable-coherence window (echo-recoverable fraction versus delay), both as functions of the swept ratio. *Null model*:

standard circuit-QED master-equation predictions for the same sweep — both observables vary smoothly, with scales set by the engineered rates, and nothing distinguished happens at any particular ratio. *Framework curve*: to be supplied by the §9 calculation; the qualitative expectation is a kink or crossover in both observables co-located at a single ratio, with width tracking w . *Confounds to be budgeted*: thermal occupation of the readout mode, drive-power dependence, non-Markovian filter response. *Status*: the null model is computable today; the framework curve is not yet — which is the honest reason this section is a direction and not a claim. Once that curve exists, the cross-platform version (two systems with matched ratio and wildly different mass, size, and particle number showing matched boundary phenomenology) becomes the discriminating form.

8.3 Macromolecule interferometry, re-read

The interference of large molecules — C_{70} , functionalized oligomers, and by 2019 a 25 kDa, 2000-atom molecule [Arndt et al. 1999; Hornberger et al. 2012; Fein et al. 2019] — is standardly narrated as a march up the mass axis toward an anticipated breakdown. The threshold account predicts the march continues indefinitely *as long as the interfering system stays above its layer*: mass is not the variable. Visibility is lost to decoherence (thermal emission, collisional which-path labeling — stage-(2) physics, fully accounted by the standard theory [Hornberger et al. 2012]) and *committed* only when a detector-grade lock occurs. The discriminating structure is recoverability: stage-(2) visibility loss is in-principle reversible (echo-type recovery; capture-without-registration, §4.1), stage-(3) loss is not. An experiment that could rephase after a putative “collapse” — recovering visibility that CSL or gravitational reduction says is gone for good — separates the programs cleanly: spontaneous-collapse theories predict a mass-scaled, unrecoverable loss channel; the threshold account predicts zero unrecoverable loss until coupling to an above-threshold system, at any mass.

8.4 Leggett–Garg tests as cut-location experiments

Macrorealism, as operationalized by the Leggett–Garg inequality [Leggett & Garg 1985; Emary et al. 2014], asks whether a system has a definite state at all times. The threshold account’s answer is sharp: definiteness is *purchased at the layer*, so LGI violations should persist for any system maintained below threshold ($\kappa_{\text{ret}} \ll K$... i.e. kept coherent, weakly monitored) regardless of macroscopicity, and disappear exactly when the monitoring chain includes an above-threshold lock. Superconducting circuits — macroscopic by particle count, atomic-grade by w (Table §3.3) — are the existing confirmation: they violate LGI robustly. The discriminator against mass-scaled programs recurs: CSL predicts LGI violations fail at a mass/size scale independent of coupling; the threshold predicts they fail at a coupling scale independent of mass. The two axes are experimentally orthogonal, and platforms exist along both.

8.5 What would falsify this paper

- (i) A system demonstrably held above its layer ($\kappa_{\text{ret}} \ll K$, autonomous phase confirmed by injection-locking phenomenology) that nonetheless shows *unrecoverable* interference loss scaling with mass or size — the CSL/OR signature — would falsify the claim that κ_{ret}/K is the only boundary variable.
- (ii) A demonstration that outcome commitment occurs in a regime where the phase is provably slaved (e.g. registration without any mode crossing its layer) would falsify the identification of the cut with the lock.
- (iii) Boundary phenomenology (deviation statistics, lock-time distributions) that fails to collapse onto the single variable κ_{ret}/K across platforms would falsify the crossover structure.
- (iv) Inherited from Paper 1: a controlled open-system calculation showing the slaved-phase statement fails under field quantization would remove this paper’s foundation — we flag it as the live theoretical risk, not a rhetorical one.

9. Discussion

Relation to the settled literature. This paper is structured as a completion, not a rival, of three bodies of work. Decoherence theory supplies stage (2) entire; we adopt its pointer bases, its timescales, and its own verdict that selection lies outside its scope. Allahverdyan–Balian–Nieuwenhuizen supply registration; their Curie–Weiss apparatus is, in our language, the machinery *below* the layer, and their explicit assumption of

a selected asymmetry is the interface our stage (3) fills. Von Neumann’s movability theorem is preserved with its domain finally drawn. The genuinely contested relationship is with the programs that also claim stage (3): against engineered-noise reduction models we claim the same phenomenological structure with the noise identified and the fairness proved rather than stipulated (Paper 1), at the price of conditionality on detector premises; against gravitational reduction we claim a mechanism and a non-gravitational, non-universal threshold — the cut is at $\kappa_{\text{ret}} \sim K$, which gravity does not set.

Open problems, in the panel’s canonical phrasings. (i) *The single continuous model* (GPT-5.6): the sub-threshold linear equation contains no K ; construct one dynamical model, derived from an explicit $H = H_{\text{field}} + H_{\text{absorber}} + H_{\text{bath}} + H_{\text{int}}$, valid on both sides of the boundary, exhibiting which mode becomes marginal at $\kappa_{\text{ret}}/K = 1$ and how the phase degree of freedom and its saturation emerge from absorber microphysics — this subsumes the κ_{ret} -ansatz derivation ([Paper 1] §9.4(viii)) and the lock’s closed dynamics in one program. (ii) *The vacuum-restored-phase question* (Grok): does a fully quantized common field restore an effective free phase below the classical threshold? — the decisive inherited risk, now stated as a precise calculation. (iii) The quantitative crossover profile: Figure 1 is a schematic oscillator analogy; the first-principles winding-rate and slip-time profiles for a physical absorber (P4(i) level structure explicit) would convert §8.2’s protocol from sketch to prediction, including the framework curve its null model currently lacks. (iv) The K - Γ prefactor for one concrete absorber (§3.2). (v) The track-cascade statistics of §4.2: Mott’s conditional stakes composed with per-grain Born games should reproduce track phenomenology; a worked calculation would make the taxonomy’s second row as quantitative as its first. (vi) The crossover’s completion in the framework paper [Paper 3]. (vii) A dedicated related-work clearance against the synergetics/laser-threshold literature (Haken school): if the qualitative “measurement as threshold crossing” claim has precedent there, the delta — the width formula, the movability-domain theorem, the detector taxonomy — must be stated against it explicitly; flagged by the internal review, not yet performed.

The one-sentence summary. The Heisenberg cut is the locking layer of the detector: free to place wherever linearity reigns — which is a theorem — and pinned, with computable width $w = K/\omega$, exactly where it does not; the classical world begins where the phase stops being slaved, and that address is different, calculable, and testable in every laboratory in physics.

Appendix A — Figure methods

The figure is a **schematic oscillator analogy** (a classical stochastic model with gain and saturation put in by hand), not a simulation of a detector; §3.1 and §9(i) state what a detector-derived version requires. Panel (a): injected Stuart–Landau oscillator $\dot{a} = (g - |a|^2)a - i\Delta a + F + \xi(t)$, $\Delta = 1.0$, $F = 0.35$, complex white noise of strength $D = 10^{-4}$; Euler–Maruyama, $dt = 0.01$; transient $T = 200$ discarded, winding accumulated over $T = 800$; 26 gain values $g \in [-1, 1.5]$. **Two boundaries, both marked:** the Hopf threshold at $g = 0$ (free phase exists) and the injection-locking boundary $g = (F/\Delta)^2 = 0.1225$ (free phase runs; overlay $\sqrt{\Delta^2 - F^2/g}$ valid only there, with limit-cycle amplitude $\sqrt{g}$ and effective locking range $K_{\text{eff}} = F/\sqrt{g}$). At the stated noise the measured running onset is $g \approx 0.35$ — the boundary is noise-smeared upward, and the reviewed draft’s caption, which placed “the threshold” at $g = 0$, conflated the two boundaries (corrected in v0.3 after the review panel’s independent re-run of this script). Convergence: below the Hopf threshold the measured winding rate is $\lesssim 4 \times 10^{-5}$ (noise floor at these D , dt , and averaging time; halving dt moves no plotted point by more than the marker size). Panel (b): Adler equation $\dot{\varphi} = \Delta\omega - K \sin \varphi$, $K = 1$; slip period measured as the time for one full 2π advance, $dt = 5 \times 10^{-4}$; 25 detunings $\Delta\omega/K \in [1.02, 5]$; analytic curve $2\pi/\sqrt{\Delta\omega^2 - K^2}$; at $\Delta\omega/K = 5$ simulation and analytic agree to 4×10^{-4} relative. Reproduction: `python3 threshold_clock.py` in `paper2_sims/` (numpy + matplotlib; seed pinned, 20260728), output `figures/p2_fig1_threshold.png`; the repository commit accompanying this revision is the immutable reference.

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